Legacy Code

Input file:	`standard input`
Output file:	`standard output`
Time limit:	8 seconds
Memory limit:	1024 megabytes

You have a piece of legacy code:

$$
\begin{array}{l}
\text{for } t = 1 \text{ to } k \text{ do} \\
\quad a_{x_1} \leftarrow (a_{u_1} \times a_{v_1}) \bmod MOD \\
\quad a_{x_2} \leftarrow (a_{u_2} \times a_{v_2}) \bmod MOD \\
\quad a_{x_3} \leftarrow (a_{u_3} \times a_{v_3}) \bmod MOD \\
\quad \vdots \\
\quad a_{x_m} \leftarrow (a_{u_m} \times a_{v_m}) \bmod MOD \\
\text{end for}
\end{array}
$$

where $MOD = 998\,244\,353$ and $x_i, u_i, v_i \in \{1, \ldots, n\}$ for $i = 1, \ldots, m$.

Given the initial values of the variables $a_1, a_2, \ldots, a_n$, compute their final values after the loop runs k times. You need to answer q such queries.

Input

The first line contains three integers n, m and q ($1 \le n \le 200$, $1 \le m \le 2 \times 10^5$, $1 \le q \le 200$).

The next m lines each contain three integers x_i, u_i, v_i ($1 \le x_i, u_i, v_i \le n$).

Each of the next q lines represents a query. Each query contains $n + 1$ integers:

- the number of iterations k ($1 \le k \le 10^6$),
- the initial values of the n variables $a_1, a_2, \ldots, a_n$ ($0 \le a_i < 998\,244\,353$).

Output

For each query, output a single line containing n integers — the final values of $a_1, a_2, \ldots, a_n$ after executing the code, each modulo $998\,244\,353$.

Example

standard input
7 8 4 3 1 2 4 3 3 5 6 7 7 1 3 2 2 2 6 6 2 1 5 6 7 7 7 1 2 1 1 1 1 1 1 2 1 2 3 4 5 6 7 3 1 1 0 1 0 1 0 100 11 45 14 19 19 8 10
standard output
1 1 2 4 1 1 16 36864 16 4032 16257024 96 384 227524445 0 1 1 1 0 1 1 244858606 337827833 584144609 343808937 412777304 279017446 926699892